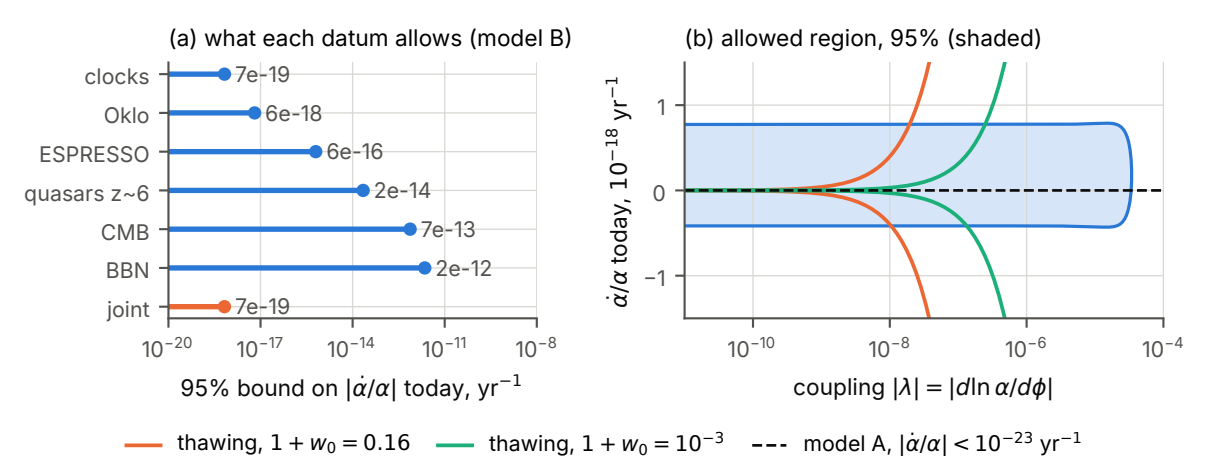
(a) what each datum allows (model B)
clocks
Oklo
ESPRESSO
quasars z~6
CMB
BBN
joint
7e-19
6e-18
6e-16
2e-14
7e-13
2e-12
7e-19
10^{-20}
10^{-17}
10^{-14}
10^{-11}
10^{-8}
95% bound on $|\dot{\alpha}/\alpha|$ today, yr^{-1}
(b) allowed region, 95% (shaded)
$\dot{\alpha}/\alpha$ today, $10^{-18}\ \mathrm{yr}^{-1}$
1
0
−1
10^{-10}
10^{-8}
10^{-6}
10^{-4}
coupling $|\lambda| = |d\ln\alpha/d\phi|$
thawing, $1 + w_0 = 0.16$
thawing, $1 + w_0 = 10^{-3}$
model A, $|\dot{\alpha}/\alpha| < 10^{-23}\ \mathrm{yr}^{-1}$